

Project Interim Report
On
Movie Ticket Booking System
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

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Abstract

The Movie Ticket Booking System is a Java-based application developed using Object-Oriented Programming (OOP) concepts. This project simulates the real-world process of booking movie tickets in a theatre. In modern cinema systems, users need a reliable platform to select seats and book tickets efficiently without errors.

This system allows users to view movie details, check seat availability, and book seats using a 2D array structure. Each seat is represented as an object with attributes such as type, price, and booking status. The system prevents double booking by validating seat availability and calculates the total cost based on selected seats.

The project demonstrates key OOP concepts such as encapsulation, object interaction, and use of arrays. Overall, it provides a simple and efficient model of a real-world ticket booking system.

Introduction

The Movie Ticket Booking System is designed to simulate how cinema booking platforms work. It helps users select seats and book tickets in an organized and error-free manner. The system is built using Object-Oriented Programming, making it modular and easy to manage.

The project mainly consists of three classes:

- Movie Class → stores movie name and show timing
- Seat Class → represents each seat with type, price, and booking status
- Booking Class → manages seat allocation and booking process

The system uses a 2D array to represent seat layout (rows and columns), making seat selection realistic and easy to understand.

Objectives

- Manage movie ticket booking process
- Prevent double booking of seats
- Track seat availability using 2D array
- Calculate total ticket cost automatically
- Apply OOP concepts in real-world scenario

Scope

This project is suitable for small-scale theatre booking systems. It can be used as a basic model for understanding how booking systems work. In the future, it can be enhanced with features like multiple movies, online payment, and graphical user interface (GUI).

System Requirements

Hardware:

- Computer/Laptop with minimum 4GB RAM

Software:

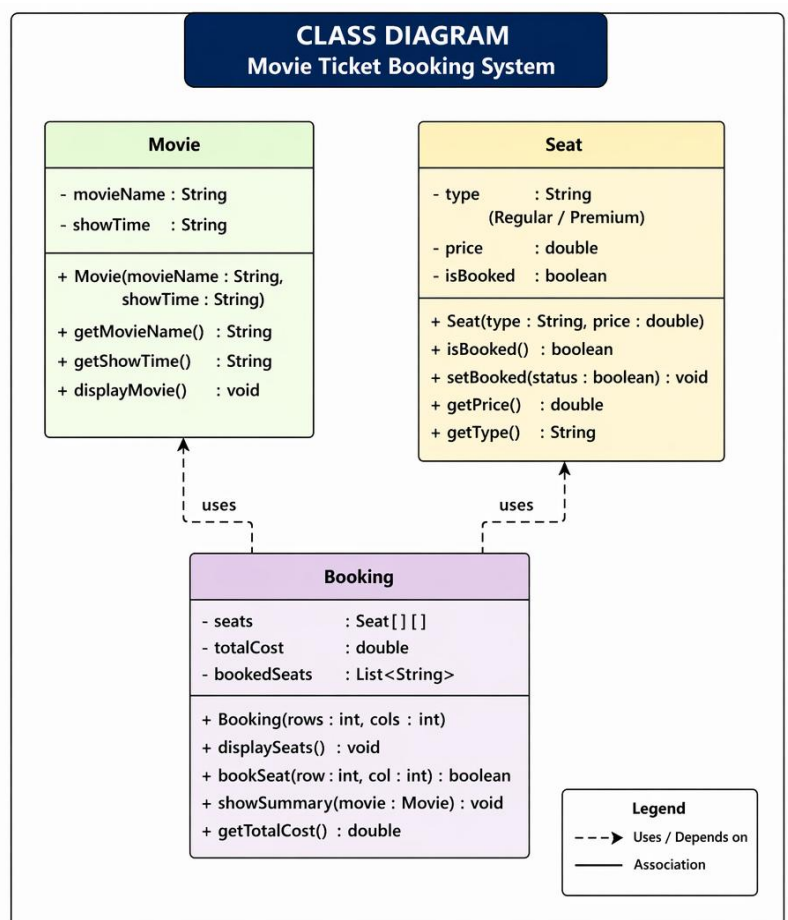
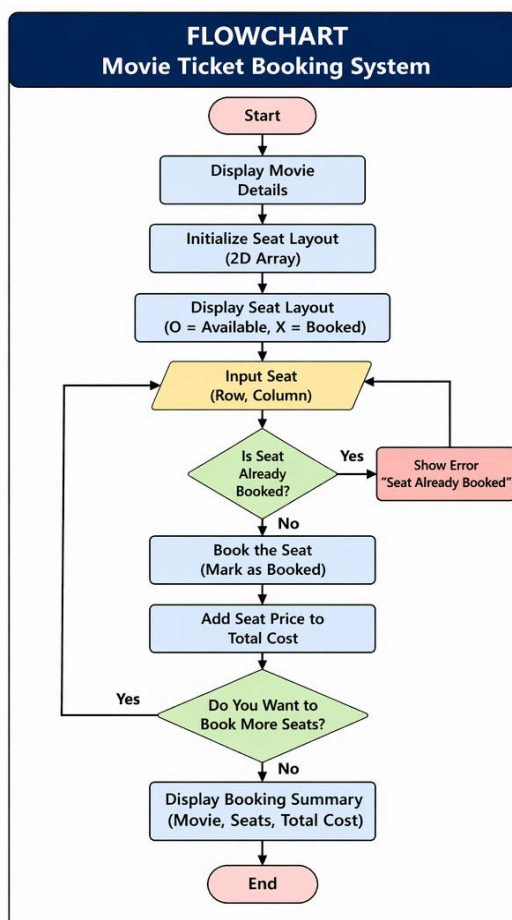
- Java JDK
- IDE (VS Code / Eclipse / IntelliJ)

Methodology

The system works step-by-step based on user interaction:

- First, a Movie object is created with movie name and show time
- A 2D array of Seat objects is initialized to represent the theatre layout
- The system displays available seats using O (available) and X (booked)
- The user selects seat position (row and column)
- The system checks if the seat is already booked
 - If yes → shows error
 - If no → confirms booking
- The seat price is added to the total cost
- The process continues until user stops booking
- Finally, booking summary is displayed

Flowchart & Class Diagram



Movie Class

Variables:

- movieName
- showTime

Methods:

- Constructor
- Getter methods

Seat Class

Variables:

- type
- price
- isBooked

Methods:

- Constructor

Booking Class

Variables:

- Seat[][] seats (2D array)
- totalCost
- bookedSeats list

Methods:

- displaySeats()
- bookSeat()
- showSummary()

Implementation

The system is implemented using three main classes:

Movie Class:

Stores movie details such as name and show time. It uses encapsulation with getter methods.

Seat Class:

Represents each seat in the theatre. Each seat has a type (Regular/Premium), price, and booking status.

Booking Class:

Handles the main logic of the system.

- Uses a 2D array to manage seats
- displaySeats() shows seat layout
- bookSeat() prevents double booking
- showSummary() displays final booking details

Sample Input/Output

Seat Layout:

O O O O

O X O O

O O O X

Selected Seats: (1,2), (2,3)

Total Cost: ₹400

Booking Confirmed!

Future Scope

- Add multiple movies and show timings
- Implement seat cancellation feature
- Add GUI using Java Swing
- Integrate online payment system
- Store data using database

Gantt Chart

Planning & Structure

The project is divided into tasks such as design, coding, testing, and documentation.

Time Management

- Design → Day 1–2
- Coding → Day 3–5
- Testing → Day 6
- Documentation → Day 7

This helps in completing the project efficiently without delay.

Conclusion

The Movie Ticket Booking System successfully demonstrates the use of Object-Oriented Programming concepts and 2D arrays. It provides an efficient way to manage seat booking while preventing errors like double booking. The project is simple, scalable, and can be extended with advanced features in the future.

References Book Format:

Book:

[1] H. Schildt, *Java: The Complete Reference*, McGraw-Hill

Website:

[2] Oracle Java Documentation – <https://docs.oracle.com>